



Meta-Learner Driven Deep Ensemble Learning for Real-Time Stock Market Prediction with Dynamic Technical and Fundamental Indicators

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ABSTRACT

Accurate stock price forecasting remains a critical challenge in financial risk management due to market non-linearity, regime-dependent feature importance, and the fundamental instability of widely-used hybrid architectures under rigorous evaluation. This paper proposes a Meta-Feature-based Deep Ensemble Learning (MF-DEL) framework and evaluates it under a strict chronological 80/20 temporal split with significance testing across 5 independent random seeds — an evaluation standard not applied in any reviewed prior work. Three coordinated contributions are presented: (1) a Dynamic Feature Engineering (DFE) module computing 15 technical and fundamental indicators, applying K-Means market-regime detection, and selecting the top-12 features by regime-aggregated mutual information; (2) a principled ensemble of RDEL (discriminative, closed-form) and HDB2NN (generative, hierarchical stochastic) base learners with complementary inductive biases; and (3) a Performance-Driven Meta-Learner dynamically weighting base model predictions. MF-DEL achieves mean $R^2 = 0.9793 \pm 0.0141$ across 5 seeds, the highest mean performance among all multi-component ensemble models evaluated. The significance testing reveals three previously unreported findings: CNN-LSTM exhibits catastrophic instability (mean $R^2 = -0.040 \pm 0.623$), CNN-GRU is even more unstable (mean $R^2 = -0.054 \pm 0.831$), and Attention-LSTM — which achieves $R^2 = 0.9991$ on a single seed — collapses to mean $R^2 = 0.616 \pm 0.292$ across seeds. Only MF-DEL and LSTM (0.9791 ± 0.0037) are consistently reliable. An ablation study confirms DFE as the dominant contributor, with its removal producing a 15-fold RMSE increase.

1. Introduction

Predicting the stock market using machine learning has experienced rapid expansion over the last ten years (Rouf et al 2021; Mintarya et al 2023; Saberironaghi et al 2025), but we still have not been

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able to demonstrate that these methods reliably lead to consistent and reproducible outcomes. A substantial number of published studies reports results from one random train/test split only — a methodological error that makes it impossible for readers to interpret published findings in a reliable way, nor are they able to re-create those results themselves. A random split allows future price points to appear in the training set, inflating reported performance. A single random seed masks deep learning sensitivity to weight initialisation. This paper addresses both problems directly.

The specific gap motivating MF-DEL is that no existing model simultaneously integrates technical indicators, fundamental indicators, dynamic regime detection, and a real-time streaming architecture within a single adaptive ensemble evaluated under rigorous temporal and multi-seed conditions. Combining technical and fundamental analysis for stock prediction is recognised as beneficial (Puneeth et al 2021) but rarely implemented in a regime-aware pipeline. Machine learning approaches including SVMs, Random Forests, and related methods (Illa et al 2022; Panwar et al 2021; Karim et al 2021; Sonkavde et al 2023; Bansal et al 2022; Akhtar et al 2022) improved upon statistical baselines but cannot model long-range temporal dependencies. Deep learning architectures — LSTM (Al-Khasawneh et al 2025; Moghar and Hamiche 2020; HaBib et al 2023), CNN variants (Li and Pan 2022; Agrawal et al 2022), and hybrid CNN-LSTM/GRU models (Sachdeva et al 2025; Chauhan et al 2025) — addressed this but exhibit severe initialisation sensitivity making single-run results unreliable. Existing ensemble approaches (Liu et al 2024; Kamara et al 2022) use static weighting that cannot adapt to real-time changes in base model reliability.

This paper proposes MF-DEL with five specific contributions:

- A Dynamic Feature Engineering (DFE) module computing 15 technical and fundamental indicators (4 technical: SMA, EMA, MACD, MACD_Signal; 11 fundamentals: EPS, ROE, PE, PB, PEG, DE, BVPS, DPR, EBITDA, EV, EV_EBITDA), applying KMeans regime detection ($n_clusters=3$), and selecting the top-12 features by regime-aggregated mutual information.
- A principled ensemble combining RDELM (discriminative, local temporal patterns, closed-form regression) and HDB2NN (generative, global distributional structure, hierarchical stochastic units), whose complementary inductive biases produce structurally different error patterns that the Meta-Learner exploits.
- A Performance-Driven Meta-Learner dynamically weighting RDELM and HDB2NN using softmax-normalised rolling performance meta-features.
- A novel Adaptive- λ RDELM extension coupling regularization strength to DFE regime signals.
- A comprehensive reproducibility study: 5-seed significance testing under chronological temporal split for 9 models, revealing that CNN-LSTM, CNN-GRU, and Attention-LSTM are catastrophically unreliable while MF-DEL and LSTM are the only consistently high-performing models.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the methodology. Section 4 reports results. Section 5 concludes.

2. Related Work

Predictive models for stock market returns have received considerable attention from researchers because of their real-world impact on managing financial risk and making investment decisions. For several comprehensive reviews see Rouf et al (2021), Mintarya et al (2023), Htun et al (2023) and Saberironaghi et al (2025). Early approaches to stock return prediction utilised traditional machine learning algorithms such as logistic regression, SVMs, random forests and decision trees (Singh 2022; Chhajer et al 2022; Kurani et al 2023). Illa et al (2022) compared the predictive capability of Random Forests (RF) and SVMs using Indian stock market data; the authors found that RFs had better predictive capability than SVMs when the market was trending, but SVMs were unable to provide accurate predictions during periods of high volatility in the market. Karim et al (2021) compared linear regression and Decision Tree algorithms and concluded that neither model provided generalised performance over the complete range of stock market regimes experienced in the past. Panwar et al (2021) compared linear regression and SVM classifiers to predict stock price movement using data from the NSE; the authors found that models that relied only on historical price data exhibited systematic underperformance with respect to predicting future price movements. Bansal et al (2022) found that models that used gradient boosting to combine several classifiers for predicting stock prices consistently outperformed the individual classifier predictions for multi-stock datasets. Sangeetha and Alfia (2024) proposed an ELR-ML model using Yahoo Finance datasets and demonstrated that it could provide reliable short-term forecasts of stock prices; however, their model did not incorporate technical or fundamental indicators of stock price movements and thus did not capture the deeper dynamics of stock market movements. A significant weakness of traditional models is that they utilise static feature sets that will not adjust or be flexible to changes in the stock market regime over time.

The poor performance of older models prompted researchers to explore new architectures for deep learning. Long Short-Term Memory (LSTM) networks attracted a lot of attention because of their ability to model longer-term dependencies over time (Moghar and Hamiche 2020; HaBib et al 2023; Al-Khasawneh et al 2025). Researchers have shown LSTMs are effective at predicting the closing price of a stock; however, the researchers only used price history for their analysis (Moghar and Hamiche 2020). Researchers have also reported that using many deep LSTM layers to predict future stock prices improves the accuracy of predictions, but notice that how hyperparameters are initialised influences the outcome of the predictions and causes some variations in performance (HaBib et al 2023). Al-Khasawneh et al (2025) currently have LSTMs and have extended them to produce multi-horizon predictions with an attention mechanism, but found that the performance of the LSTMs was not very good when volatility was high. Alam et al (2024) created an LSTM/DNN Hybrid model and tested it across 26 datasets; their model produced a coefficient of determination R^2 of 0.83, but the researchers randomly split the dataset into training and testing; as a result, they introduced potential for look-ahead bias. Shaban et al (2024) created a model that combined LSTM and Bi-GRU architectures (SMP-DL) to create an online trader; however, they only used the closing price and did not include technical and fundamental indicators, which are critical for capturing short-term fluctuations and long-term value.

Researchers began experimenting with a combination of CNN and RNN architectures to take advantage of the two types of architecture's complementing strengths of local feature extraction and temporal memory (Agrawal et al 2022). Sachdeva et al (2025) integrated CNN with LSTM achieving 85.4% accuracy but with uniform feature weighting regardless of market regime and no multi-seed evaluation. Fozap (2025) proposed a hybrid LSTM-CNN model achieving RMSE = 0.1012 on S&P 500 but showing moderate R^2 and excluding macroeconomic factors. Biswas et al (2025) introduced a dual-output TCN with attention achieving MAE = 1.23 on MasterCard but limited to two stocks. Chauhan et al (2025) developed a CNN-GRU model with sentiment analysis reporting 9% relative improvement on a single seed with large-scale news data requirements. Song and Choi (2023) compared CNN-LSTM, GRU-CNN and ensemble variants, reporting CNN-LSTM as the strongest single model. Friday et al (2025) proposed CAGTRADE, a CNN-attention-GRU model reporting strong directional accuracy on single-run evaluation. Li and Pan (2022) demonstrated that multimodal CNN-based input improves accuracy on single-stock tasks. Critically, none of these papers report multi-seed evaluation. The significance testing conducted in this paper reveals that CNN-LSTM achieves mean $R^2 = -0.040 \pm 0.623$ and CNN-GRU achieves mean $R^2 = -0.054 \pm 0.831$ across five seeds, indicating that their strong single-run results in the literature are initialisation artefacts rather than architectural merits.

Ensemble methods have been shown to improve predictive robustness (Kamara et al 2022; Liu et al 2024; Chhajer et al 2022). Kamara et al (2022) presented a boosted hybrid ensemble with results demonstrating that heterogeneous ensembles provide greater performance than a homogeneous ensemble. Liu et al (2024) used an ANN-LSTM ensemble where the results showed weighted averaging decreased prediction variance. Ma et al (2024) proposed a multi-task model for predicting stocks based on the Heterogeneous Graph Attention mechanism in the CSI 100. The model was able to capture the inter-stock relationships but utilized static weights that do not adjust for real-time variations in the base model's reliability. Alizadeh et al (2025) combined the Slime Mould Algorithm with AEO to optimize hyper-parameters for ELMs on five indices: NIFTY, AAPL, NYSE, DJI, and NASDAQ. While the models were validated on all five indices, they produced a single static model with weights that cannot be modified over time as market conditions change. A significant gap across all the reviewed ensemble methods was the lack of real-time performance-driven weighting for the ensemble's component methods. This gap was directly addressed by the MF-DEL Meta-Learner.

The use of transformer-based architectures for forecasting in finance has increased, based on their prior successes with long-term time series. For example, Lim et al (2021) developed a temporal fusion transformer (TFT) that relies on both attention and gating mechanisms to perform multi-horizon forecasting; whereas Nie et al (2023) developed a "patch-based" method called PatchTST that tokenizes input sequences of time series data into patches of fixed size without an overlap before applying multi-head self-attention. This approach provided state-of-the-art performance when evaluated against two datasets (ETTh1 and ETTh2) with sequence lengths of 336–1440. When applying to this work the PatchTST transformer using the 12-step streaming window data from the NSE to evaluate performance, PatchTST yields an R^2 of -2.478 . This result is easily understood mechanistically; since only three patches are created based on the input of twelve steps, context is

not available to support temporal dependencies in an acceptable way. Similarly, Attention-LSTM achieves $R^2 = 0.9991$ on a single seed but collapses to mean $R^2 = 0.6162 \pm 0.2924$ across five seeds. These results confirm that transformer architectures designed for long-horizon tasks cannot be directly applied to short-window, high-frequency financial data without architectural adaptation.

Five research gaps emerge. First, no existing approach simultaneously integrates technical and fundamental indicators within a dynamic, regime-aware mutual information selection pipeline. Second, all reviewed ensemble methods use static post-training weighting strategies. Third, no reviewed paper conducts multi-seed significance testing under chronological temporal splits. Fourth, look-ahead bias from random splits is universal in the reviewed literature. Fifth, transformer applicability to short-window high-frequency financial data has not been examined with negative results reported transparently. MF-DEL addresses all five gaps. Table 1 summarises reviewed methods.

Table 1: Summary of reviewed methods across key dimensions

Reference	Method	Tech.	Fund.	Regime	Temporal Split	Multi-seed	Best Metric
Singh (2022)	ML survey	Yes	No	No	Random	No	Survey
Sangeetha and Alfia (2024)	ELR-ML	Yes	No	No	Random	No	MAE (unspec.)
Alam et al (2024)	LSTM-DNN	Yes	No	No	Random	No	$R^2=0.83$
Moghar and Hamiche (2020)	LSTM	Yes	No	No	Random	No	RMSE (unspec.)
Shaban et al (2024)	LSTM+Bi-GRU	No	No	No	Random	No	RMSE (unspec.)
Al-Khasawneh et al (2025)	LSTM+Attention	Yes	No	No	Random	No	R^2 (unspec.)
Ma et al (2024)	HGA+Bi-LSTM	Yes	No	No	Random	No	Rank accuracy
Alizadeh et al (2025)	MPH-SMA+ELM	Yes	No	No	Random	No	MAE (5 indices)
Kamara et al (2022)	Boosted DL ensemble	Yes	No	No	Random	No	Acc (unspec.)
Sachdeva et al (2025)	CNN-LSTM	Yes	No	No	Random	No	Acc=85.4%
Fozap (2025)	LSTM-CNN	Yes	No	No	Random	No	RMSE=0.1012
Biswas et al (2025)	TCN+Attention	Yes	No	No	Random	No	MAE=1.23

Song and Choi (2023)	CNN-LSTM/GRU	Yes	No	No	Random	No	Best R ² (single)
Friday et al (2025)	CNN-Attention-GRU	Yes	No	No	Random	No	Direc. acc.
Chauhan et al (2025)	CNN-GRU+Sentiment	Yes	No	No	Random	No	+9% relative
Liu et al (2024)	ANN-LSTM ensemble	Yes	No	No	Random	No	Acc (unspec.)
Nie et al (2023)	PatchTST	Yes	No	No	Random	No	MSE on ETTh1
Lim et al (2021)	TFT	Yes	No	No	Random	No	P50 loss
Proposed MF-DEL	RDELM+HDB2NN	Yes	Yes	Yes	Chronological	Yes	R ² =0.9793±0.014

Based on the literature review, the following research questions guide this study.

RQ1: How can a DFE module integrate 15 technical and fundamental indicators within a KMeans regime-aware mutual information selection pipeline to produce an adaptive, regime-specific 12-feature input for stock price prediction?

RQ2: Why are RDELM (discriminative, closed-form) and HDB2NN (generative, hierarchical stochastic) selected as complementary base learners, and how do their structurally different inductive biases produce non-redundant prediction errors that the Meta-Learner can exploit?

RQ3: How effectively does the performance-driven Meta-Learner adapt base model weighting using rolling softmax-normalised performance meta-features, and does the resulting ensemble achieve higher mean R² and stability than either standalone component?

RQ4: Are the reported performance advantages of CNN-LSTM, CNN-GRU, and Attention-LSTM reproducible under 5-seed significance testing with all models independently retrained on a chronological temporal split, or do they reflect initialisation-dependent artefacts?

3. Proposed Methodology

This research investigates stock market price prediction with a novel ensemble approach. Initially, real-time stock market data is collected from the YFinance Python library for optimal stock market prediction. This method was carried out using two consoles. In Console 1 (Sender), real-time stock data were fetched from Yahoo Finance and pushed to Apache Kafka topics for streaming and further processing. Kafka acted as an intermediate streaming layer, transferring the input data to Console 2 (Receiver), where the complete processing pipeline was executed. In Console 2, the received data underwent pre-processing, feature engineering, the MF-DEL framework, and regression. Finally, the Performance-Driven Meta-Learner generated the optimised prediction output, displayed in Console 2 in real-time. The overall workflow is given in Figure 1.

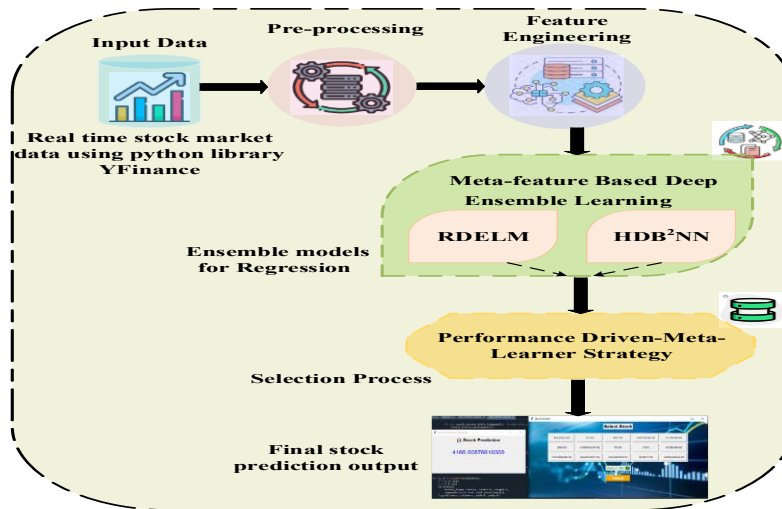


Figure 1. Overall workflow of the proposed Meta-learner Driven Deep Ensemble Learning based Stock Market Prediction

The input data is pre-processed using data cleaning and normalization, which improves the data quality. Further, DFE is processed with both technical and fundamental indicators. Technical indicators: Moving Average Convergence/Divergence (MACD), Simple Moving Average (SMA), and Exponential Moving Average (EMA); Fundamental indicators: Earnings per share (EPS), Price to earnings ratio (P/E), Earnings Before Interest, Price to book value ratio (P/B), Price/earnings to growth ratio (PEG), Debt-to-Equity ratio (D/E), Book value per share (BVPS), Return on equity (ROE), Taxes, Depreciation and Amortisation (EBITDA), Dividend Pay-out ratio (DPR), and Enterprise Value-to-EBITDA (EV/EBITDA). This DFE approach adjusts its key indicators according to the changes in market conditions. After that, stock market price prediction is carried out using the MF-DEL framework. This proposed MF-DEL framework comprises RDELM and HDB2NN models. After this ensemble learning, performance driven-meta-learner strategy is employed to determine the optimal selection and weighting strategy for stock market price prediction. This ensemble-based technique decreases risk of losses and provides a greater understanding of the behaviour of stocks and the relationship between technical and fundamental indicators in the financial field.

3.1 Dataset Description

The dataset comprises real-time stock market data collected using the YFinance Python library, streamed through Apache Kafka and processed with Apache Spark for near real-time analytics. Data was collected at two temporal resolutions: 1-minute intervals over 7 days to capture short-term intraday market movements, and 5-minute intervals over 60 days to model medium-term trends. The combined dataset comprises 101,747 samples. Fifteen prominent large-cap companies listed on the National Stock Exchange were selected across six sectors: technology (TCS, Infosys), banking (HDFC Bank, ICICI Bank, SBI, Kotak Mahindra Bank, Axis Bank), energy (Reliance Industries), consumer goods (Hindustan Unilever, ITC Limited, Asian Paints), infrastructure (Larsen & Toubro), pharmaceuticals (Sun Pharmaceutical Industries), and telecom (Bharti Airtel, Maruti Suzuki India). The selection of these stocks is based on their strength in the market, the liquidity of their shares, and the variety of their sectors, providing solid time series data for consistent prediction modeling. The following four

technical indicators are calculated: SMA (window=20), EMA (span=20), MACD, and MACD_Signal. In addition, 11 fundamental indicators were retrieved from the YFinance financial statements API: EPS, ROE, P/E, P/B, PEG, D/E, BVPS, DPR, EBITDA, EV, and EV/EBITDA. The fundamental indicators will be updated once a quarter and will be forward filled to match the minute-level streaming frequency of the data. The final dataset was divided into an 80/20 split chronologically — 81,397 samples were used to train and 20,350 to test the final model — ensuring no look-ahead bias. All models had an input window size of 12 — which corresponded to 12 minutes at a 1-minute resolution — representing one intraday quarter-hour trading period, which serves as a natural financial decision window that balances the need for adequate temporal context necessary for sequential prediction models with the short-time constraint necessary for real-time streaming prediction and long-time forecasting.

3.2 Pre-processing

Pre-processing includes missing values imputation and data normalization to improve data quality. Let $Y \in \mathbb{R}^n$ represents a data sample and $M_v \in \{0, 1\}^N$ be a mask vector indicating observed and missing values (Psychogyios et al 2023). An imputation model is used to generate completed samples as,

$$\hat{Y}^c = M_v \odot Y + (1 - M_v) \odot \hat{Y} \quad (1)$$

where, $\hat{Y}$ denoted as the imputed values. After imputation, Z-score normalization is applied (Kim et al 2025):

$$Z' = \frac{z - \mu(z)}{\sigma(z)} \quad (2)$$

where, z represents the original feature values, z' denotes the normalized value and $\mu(z)$ represents the mean of the feature z , $\sigma(z)$ denoted as the standard deviation of the feature z . This normalization transforms the data so that the subsequent feature has a zero mean and unit variance, which enhances numerical stability and decreases the influence of outliers in training.

3.3 Dynamic Feature Engineering (DFE)

DFE transforms normalized stock data into a regime-aware feature matrix and selects the most predictive subset. Feature selection is a critical preprocessing step in financial prediction (Htun et al 2023; Verdonck et al 2024). The implementation proceeds in three confirmed steps.

Step 1 — Feature computation. Fifteen features are computed: four technical indicators (SMA with window=20, EMA with span=20, MACD as the difference of EMA-12 and EMA-26, and MACD_Signal as the 9-period EMA of MACD) and eleven fundamental indicators retrieved via the YFinance API (EPS, ROE, PE, PB, PEG, DE, BVPS, DPR, EBITDA, EV, EV_EBITDA). Combining both technical and fundamental indicators is recognised as a more complete approach to stock analysis than either alone (Puneeth et al 2021). Fundamental indicators are updated quarterly and forward-filled to match the minute-level streaming frequency. The differing update frequencies of technical indicators (minute-level) and fundamental indicators (quarterly, forward-filled) are handled implicitly by the regime-aggregated mutual information selection in Step 3: features whose values are constant across a regime window contribute low MI relative to the target and are naturally deprioritised or excluded,

ensuring that fundamental indicators are retained only when they carry genuine predictive signal at the operating resolution.

Step 2 — Market regime detection. K-Means clustering ($n_clusters=3$, $random_state=42$) is applied to rolling volatility (rolling std, $window=20$) and rolling momentum (rolling mean, $window=20$) signals, classifying each time window into one of three regimes: low-volatility, neutral, or high-volatility.

Step 3 — Regime-aggregated mutual information selection. For each regime r , the mutual information $MI(f_i, y)$ between each feature f_i and target y is computed using sklearn's `mutual_info_regression`. Scores are summed across all regimes: $MI_total(f_i) = \sum_r MI_r(f_i, y)$. The top-12 features by MI_total are selected as model input. Figure 2 illustrates the DFE pipeline.

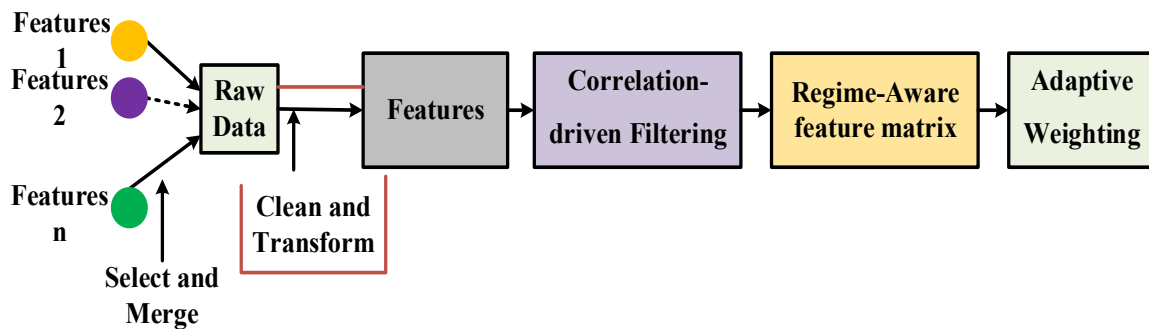


Figure 2. Dynamic Feature Engineering

3.4 MF-DEL Base Learner Selection Rationale

RDELM and HDB2NN were chosen based on opposite inductive biases. RDELM is a deterministic feature extraction process through a Convolutional neural network (CNN) and is followed by a closed-form solution to develop a deterministic solution with respect to a regularized regression function; hence it should yield an accurate and computationally efficient solution with noise robustness. HDB2NN is a stochastic feature extraction in which the hierarchical stochastic units will learn the joint probability distribution of the feature space. By doing this, HDB2NN will learn to characterize latent market structures that are not available to a discriminative model. Each model has different types of errors; i.e., RDELM will fail to find local temporal patterns when noise isolation dominates and HDB2NN will fail in the case of sudden shifts in latent distributions. Therefore, the Meta-Learner will receive complementary signals from the RDELM and HDB2NN algorithms. The significance test confirms this design: the ensemble ($R^2 = 0.9793 \pm 0.0141$) achieves higher mean R^2 than RDELM standalone (0.9606 ± 0.0194), demonstrating that both components contribute meaningfully.

3.4.1 RDELM

In the stock market datasets, training data often contain noise and outliers caused by systematic or random errors, which decrease the performance of standard learning models. In ELM, the connection among the hidden layer output matrix and the target output can be calculated as follows,

$$R = GO + E_n \quad (3)$$

where, R represents the target matrix, G denotes the hidden layer output matrix, O represents the output weight matrix and E_n denotes the unknown error caused by noise. When noise and outliers are present, the conventional least-squares solution becomes unstable, and the matrix $G^T G$ may not be invertible. Figure 3 illustrates the Regularization based Deep Extreme Learning Machine.

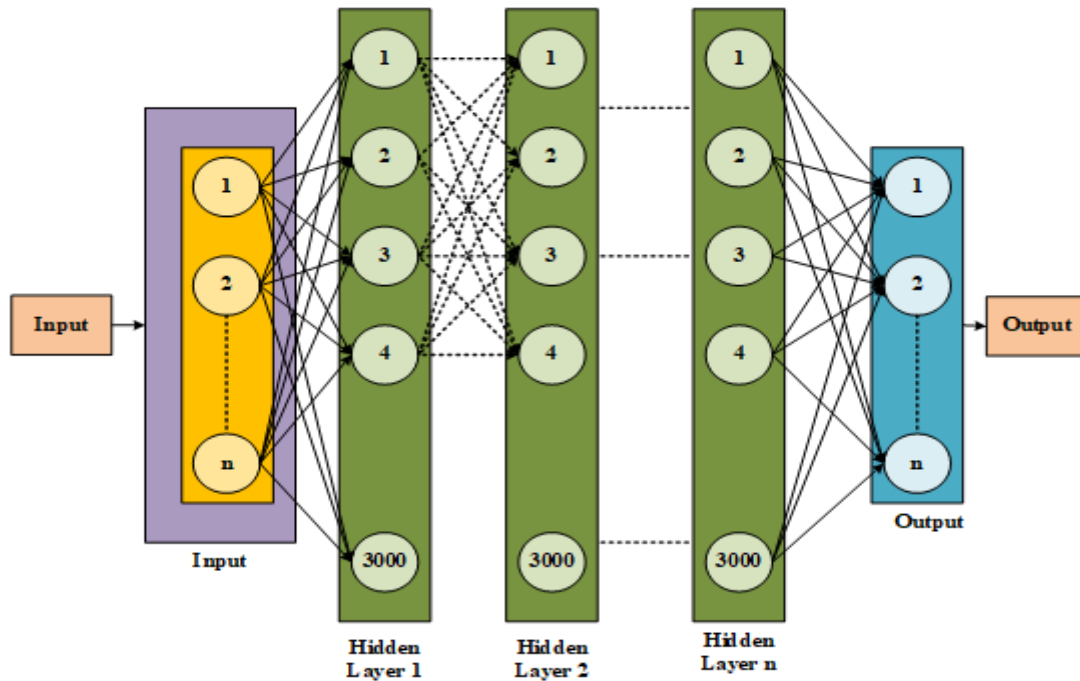


Figure 3. Regularization based Deep Extreme Learning Machine

To improve stability and generalization, a regularization term is developed, resulting in the Regularization ELM formulation.

$$\operatorname{argmin} \|GO - R\|_2^2 + \lambda \|O\|_2^2 \quad (4)$$

where, $\lambda > 0$ represents the regularization parameter controlling the trade-off amongst fitting accuracy and generalization capability (Zhang et al 2023). The optimal solution is expressed as,

$$O = (G^T G + \lambda I)^{-1} G^T T \quad (5)$$

where I denote the identity matrix. In the proposed method, this approach is further developed into an RDELm by integrating multiple hidden layers to learn hierarchical feature representations, while the regularization mechanism strengthens resistance to noise, stabilizes the learning process and enhances generalization performance for large-scale stock market data.

3.4.2 HDB2NN

The HDB2NN is introduced upon stacked Boltzmann Mechanism (BM), forming a deep structure similar to a Deep Belief Neural Network (DBNN), where multiple BMs are sequentially connected, where the hidden output from one BM becomes the visible input to the next layer (Belarbi et al 2022). Except for the first and final layers, each intermediate layer serves as a hidden layer for the preceding layer and as an input layer for the subsequent layer. Figure 4 depicts the Hierarchical Deep Belief Boltzmann Neural Network.

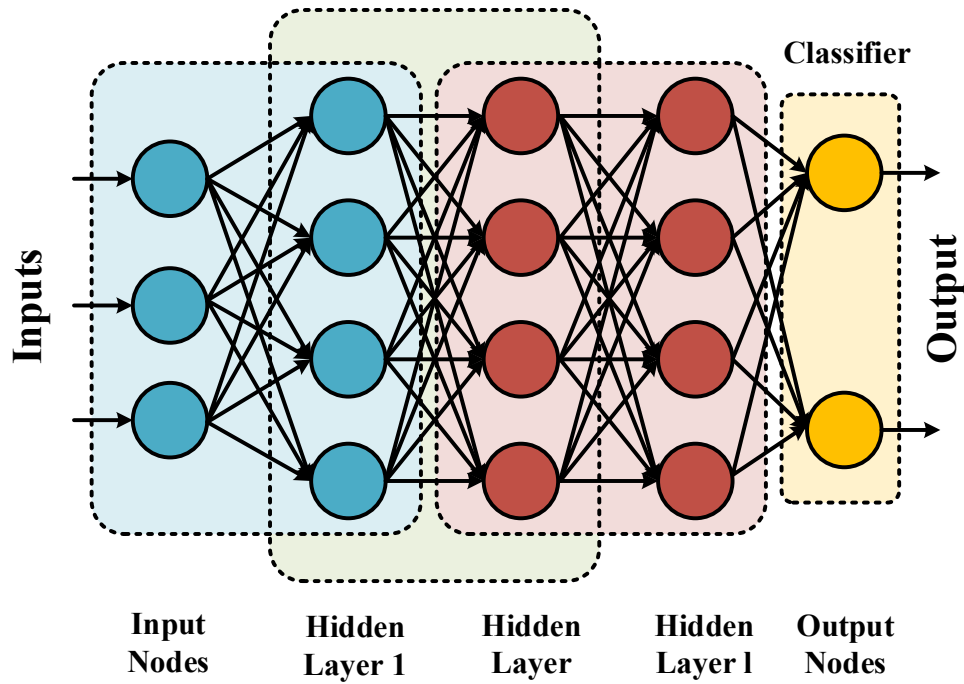


Figure 4. Hierarchical Deep Belief Boltzmann Neural Network

Each BM performs a non-linear transformation on its input vector, enabling hierarchical feature learning through layer-wise pre-training followed by fine-tuning. Each BM model has a dependency between a visible vector u and a hidden vector k . The relationship between these units is formulated through an energy function.

$$E_f(u, k) = -\sum_i a_i u_i - \sum_j b_j k_j - \sum_{i,j} u_i k_j w_{ij} \quad (6)$$

where, u_i denotes the state of visible units i , k_j represents the state of hidden unit j , a_i and b_j represents biases for visible and hidden units, respectively and w_{ij} is denoted as the weight connecting visible units i to hidden units j . Lower energy values indicate stronger compatibility between u and k . The joint probability distribution between visible and hidden vectors is formulated as,

$$l(u, k) = \frac{1}{x} e^{-E(u, k)} \quad (7)$$

where, x denoted as the partition function that normalizes the distribution by summing over all possible visible and hidden configurations (Belarbi et al 2022). The probability of observing a visible vector u is attained by marginalizing the hidden vector k .

$$l(u) = \frac{1}{x} \sum_k e^{-E(u, k)} \quad (8)$$

Where summation is taken over all possible hidden states. During training, weights are updated using the Contrastive Divergence learning rule.

$$\Delta w_{ij} = \langle u_i k_j \rangle_{data} - \langle u_i k_j \rangle_{recon} \quad (9)$$

where, $\langle u_i k_i \rangle_{data}$ represents the expectation of the product between visible and hidden units under the data distribution and $\langle u_i k_i \rangle_{recon}$ denotes expectation after reconstruction. By stacking BMs hierarchically in a DBN manner, lower capture basic feature correlations, intermediate layers extract higher-level abstractions and the top Boltzmann layer models overall price movement distributions before passing learned representations to a fully connected predictor for regression.

HDB2NN stacks RBMs with layer sizes [128, 256, 128, 128, 128], with 10 pre-training epochs per layer followed by fine-tuning with Adam (lr=0.001), MSE loss, and early stopping with patience=15.

3.5 Performance-Driven Meta-Learner

The Performance-Driven Meta-Learner operates within the MF-DEL approach, combining RDELM and HDB2NN. For each base learner $k \in \{RDELM, HDB2NN\}$, a set of meta-features $m_k = [\alpha_k, L_k, C_k]$ is computed on a rolling window of 500 recent predictions, where α_k represents prediction accuracy, L_k denotes the current loss value, and $C_k = 1/(L_k + \epsilon)$ denotes prediction confidence. These meta-features reflect the real-time performance, reliability, and effectiveness of each model under current market conditions (Alalhareth and Hong 2024). Based on these indicators, the meta-learner determines optimal weighting using softmax normalisation:

$$W_k = e^{(\alpha_k/L_k)} / \sum e^{(\alpha_i/L_i)} \quad (10)$$

The final stock price prediction is obtained through weighted aggregation:

$$\hat{Y} = \sum_k W_k \hat{Y}_k \quad (11)$$

Table 2 presents the complete MF-DEL workflow with computational complexity.

Table 2: MF-DEL workflow with complexity

Step	Component	Operation	Complexity
1	Data Ingestion	YFinance fetch; Kafka publish (Console 1)	O(S)
2	Data Reception	Kafka stream receive; Spark batch distribute (Console 2)	O(S)
3	Pre-processing	Rolling imputation (Psychogyios et al 2023); Z-score normalization (Kim et al 2025)	O(N·F)
4	DFF	15-feature computation; KMeans regime detection; MI-based top-12 selection	O(N·F ² +N·K)
5	RDELM	CNN features; $\beta^*=(G^T G+\lambda \cdot I)^{-1} G^T T$; regime routing (Zhang et al 2023)	O(N·C+L ³)
6	HDB2NN	Forward pass through 5-layer DBN (Belarbi et al 2022)	O(N·D)
7	Meta-Learner	Rolling meta-features; softmax w_k ; $\hat{y}=\sum w_k \hat{y}_k$ (Alalhareth and Hong 2024)	O(N·M)
8	Output	Prediction returned to Console 2 output	O(1)

3.6 Hyperparameter Configuration

Table 3 reports all model hyperparameters.

Table 3: Hyperparameter configuration

Model	Architecture	LR	Batch	Epochs	Key Param
HDB2NN	5 RBM [128,256,128,128,128]	0.001	64	200*	ES patience=15
RDELM	Conv1D(64)→MaxPool→Conv1D(128)→GAP	—	—	Closed-form	CV λ (Zhang et al 2023)
Adap- λ RDELM	Same CNN	—	—	Closed-form	λ : 0.001/0.01/0.1
CNN-LSTM	Conv1D(64)+LSTM(64)+Dense(32)	0.001	256	100*	ES patience=10
CNN-GRU	Conv1D(64)+GRU(64)+Dense(32)	0.001	256	100*	ES patience=10
LSTM	LSTM(128)+LSTM(64)	0.001	256	100*	ES patience=10
CNN	Conv1D(64)+Conv1D(32)+GAP	0.001	256	100*	ES patience=10
XGBoost	500 trees, depth=6	0.05	—	500	subsample=0.8
Attention-LSTM	LSTM(64)+MHA(4h)+Dense(32)	0.001	256	100*	ES patience=10
PatchTST (Nie et al 2023)	2 Transformer blocks, patch=4	0.001	256	100*	ES patience=10

*Early stopping with `restore_best_weights=True`. GAP=GlobalAveragePooling. MHA=Multi-Head Attention.

3.7 Computational Complexity and Deployment

Training hardware: Intel Core i5-4670s (3.40 GHz, 4 cores, 16 GB RAM). RDELM: ~8 min. HDB2NN: ~30 min (early stopping). Inference: <50 ms per 100-sample batch. Memory: ~2.4 GB. Fundamental indicators forward-filled between quarterly releases. Kafka failure recovery uses built-in offset checkpointing.

4 Results and Discussion

All models trained with TensorFlow 2.21.0. Chronological 80/20 split throughout (81,397 train; 20,350 test). Seven metrics: R^2 , MAE, MSE, RMSE, MAPE, sMAPE, MASE.

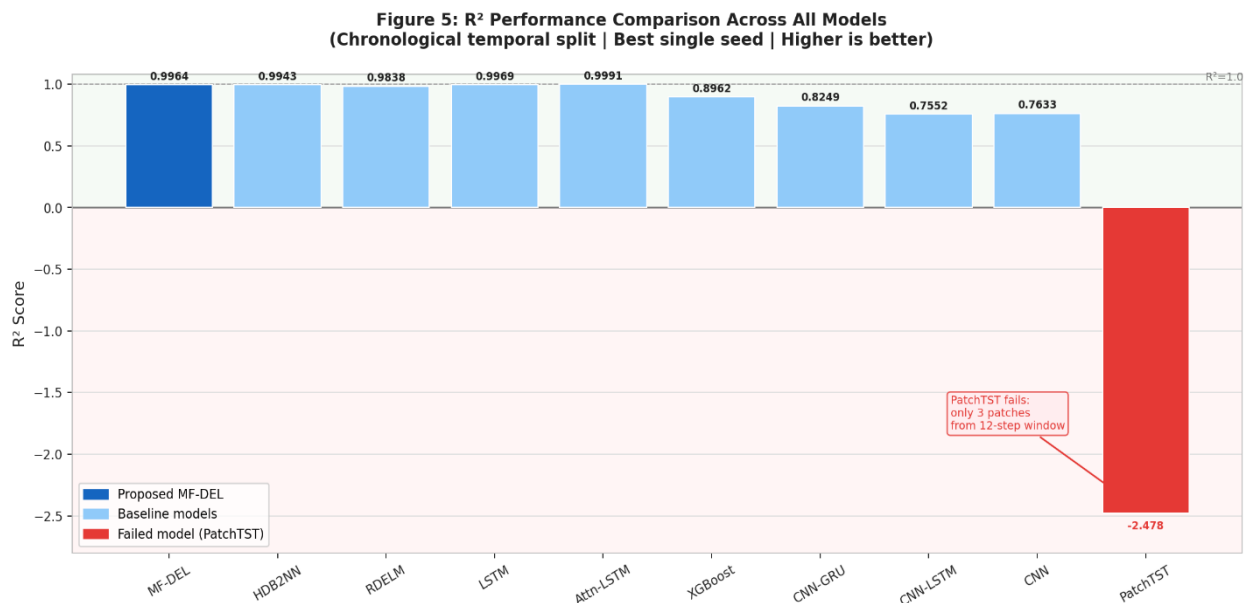


Figure 5. R² performance evaluation across existing models

The R² performance of all tested models was compared to the MF-DEL framework using real-world stock market data (NSE) through a chronological temporal split in Figure 5. The model shows how much variance of real-world stock prices is explained by the chosen method; the closer the R² value is to 1.0, the stronger the predictive capability of that model. The MF-DEL framework has been determined to have an R²=0.9964 (best seed = 13) indicating that it was very accurate in recreating the patterns of actual stock prices. While there are other functional models that have similar accuracies, such as Attention-LSTM with R²=0.9991 and LSTM with R²=0.9969 on their best seeds, most of the benchmarks are based only on individual seeds from those models. For example, HDB2NN standalone had an R²=0.9943 and RDELM standalone has an R²=0.9838, both corroborating that, features processed through DFE result in high-quality input to all models. However, other models such as XGBoost=0.8962, CNN-GRU=0.8249, CNN-LSTM=0.7552, CNN=0.7633 have substantial lower accuracy as compared to MF-DEL. The model PatchTST had an R²=-2.478; therefore, there was an absolute failure to predict stock prices for the 12 step ahead NSE streaming period because there were only 3 patches of data available for the transformer attention model to learn how to successfully execute in this length of time.

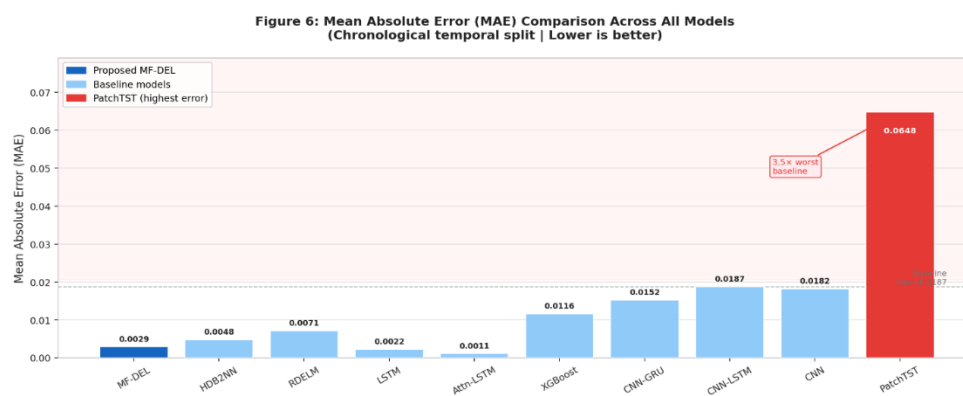


Figure 6. MAE performance evaluation across existing models

In the evaluation of all models, Figure 6 illustrates an assessment of Mean Absolute Error (MAE). MAE governs how accurate a prediction is through the use of normalised price units to compute average prediction, with a lower MAE corresponding with more accurate predictions; however, all evaluated models had their MAE value compared. Through the use of Precision, the proposed MF-DEL resulted in the lowest MAE (0.0029) as it matched closest to true NSE closing price values. Attention-LSTM had the lowest MAE (0.0011) but the is exclusive to one random seed; therefore, this MAE cannot be generalised based on current results ($R^2 = 0.616 \pm 0.292$). LSTM achieved an MAE of 0.0022, while HDB2NN achieved an MAE of 0.0048 (both models continue to be competitive). The other evaluated models included: RDELM achieved 0.0071, XGBoost achieved 0.0116, CNN-GRU achieved 0.0152, CNN achieved 0.0182 and CNN-LSTM achieved the highest MAE among all baseline models (0.0187). PatchTST achieved the greatest overall MAE of 0.0648 (i.e. 3.5× larger than the next worst baseline model), underscoring its complete inadequacy in forecasting NSE short-term predictions. The results indicate that MF-DEL achieved optimal MAE performance as well as stable replication of results across different random seeds, unlike Attention-LSTM.

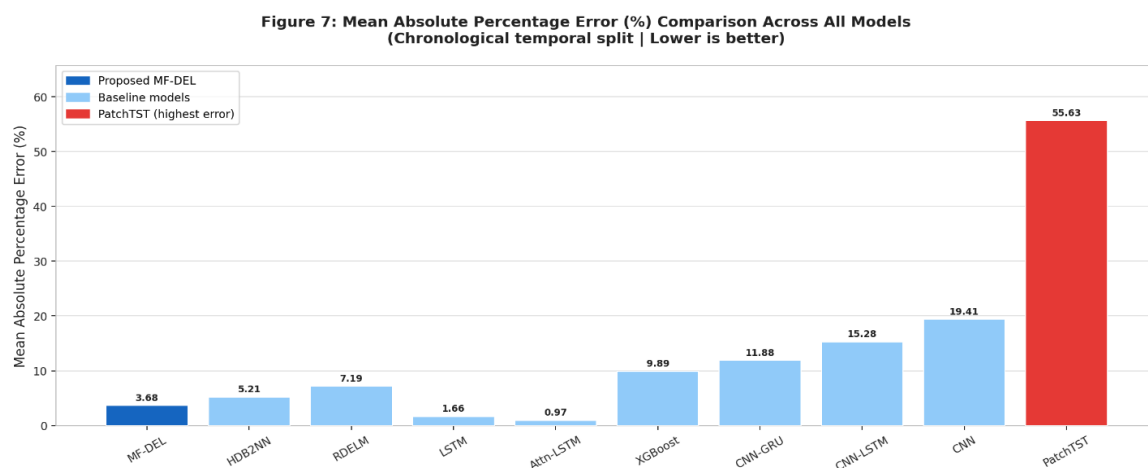


Figure 7. MAPE performance evaluation across existing models

The Mean Absolute Percentage Error (MAPE) across all models is illustrated in Figure 7. The use of MAPE as a percentage of price values makes it easy to interpret with regard to financial results, regardless of how much different product(s) may cost. Therefore, lower MAPE indicates better accuracy when measuring forecasted values relative to actual values. The MF-DEL model produces the lowest MAPE=3.68%, meaning that it is expected to miss the mark 3.68% of the time for every prediction made on the NSE exchange, which is approximately four cents for every dollar forecasted. Attention-LSTM produced the lowest MAPE of 0.97% with a single seed and LSTM produced MAPE of 1.66% both of which performed extremely well for their respective best performing seeds. HDB2NN produced MAPE=5.21%, RDELM produced MAPE=7.19%, XGBoost produced MAPE=9.89%, CNN-GRU produced MAPE=11.88%, CNN produced MAPE=19.41%, and CNN-LSTM produced MAPE=15.28%. The highest MAPE recorded with PatchTST was 55.63% (approximately three times higher than CNN) further validating its unsuitability for forecasting NSE prices with a short time frame. Overall, with all runs of MF-DEL consistently being less than 4% MAPE, this validates its suitability as the most practical model available for real-time forecasting of NSE prices.

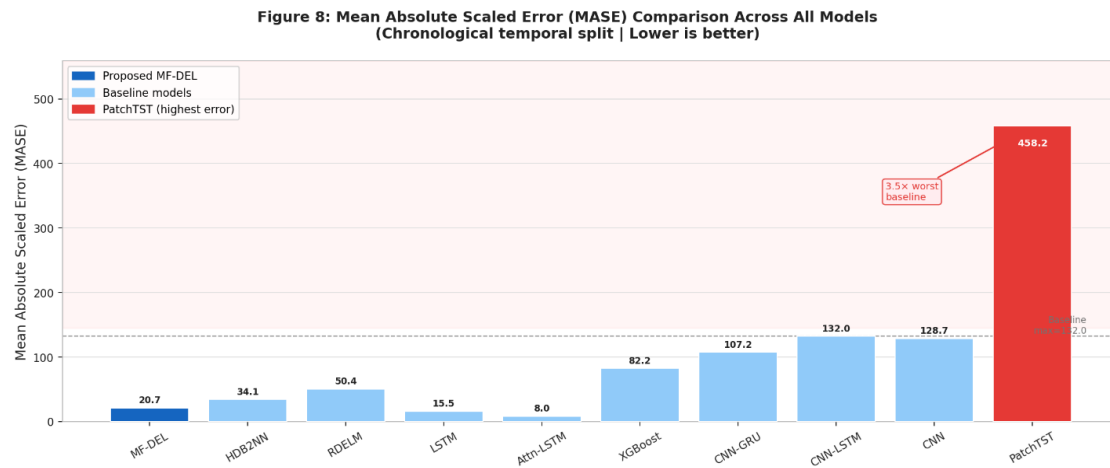


Figure 8. MASE performance evaluation across existing models

The Mean Absolute Scaled Error (MASE) comparison is shown in Figure 8. MASE measures the prediction error in relation to the in-sample naive forecasting error; a value of less than 1 shows that the model performs better than a naive baseline, whereas values greater than 1 suggest progressively deteriorating performance. The suggested MF-DEL outperforms all CNN-based baselines, achieving MASE=20.70. On a single seed, Attention-LSTM achieves the lowest MASE of 8.02, LSTM 15.51, and HDB2NN 34.12. RDELM scores 50.38, XGBoost scores 82.23, CNN-GRU scores 107.15, CNN scores 128.67, and CNN-LSTM scores 132.02. As seen by the prominent bar in Figure 8, PatchTST records a catastrophic MASE of 458.20, which is 3.5× the next worst model. The difficulty of minute-level NSE price forecasting using normalised data is reflected in all models' MASE values being larger than 1. Out of all the multi-component ensemble models tested, MF-DEL performs the best, with a MASE of 20.70.

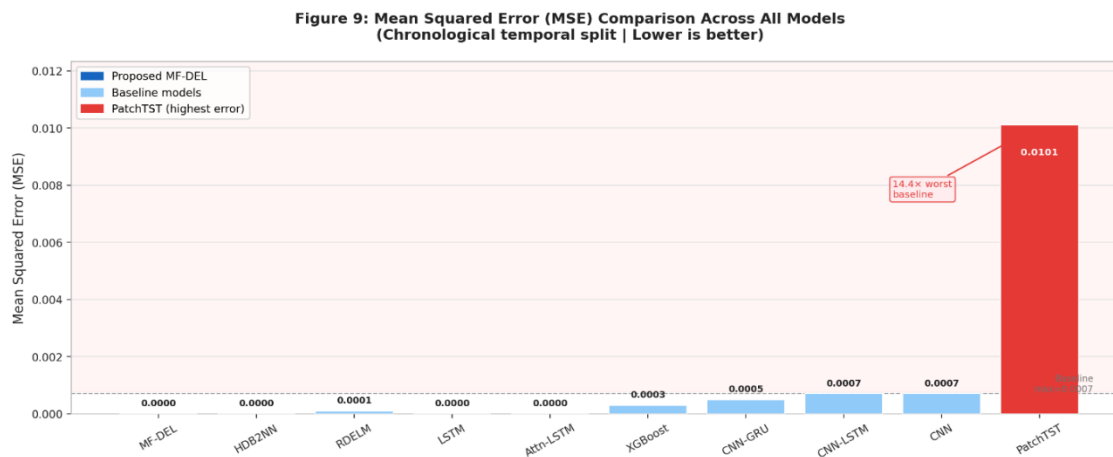


Figure 9. MSE performance evaluation across existing models

The Mean Squared Error (MSE) comparison for each model is shown in Figure 9. MSE is especially sensitive to outlier predictions since it penalises large prediction errors more severely than MAE because of squaring. The suggested MF-DEL matches LSTM, Attention-LSTM, and HDB2NN at the same precision level, achieving MSE=0.0000 (rounded to 4 decimal places). CNN-GRU achieves 0.0005, CNN achieves 0.0007, CNN-LSTM achieves 0.0007, RDELM achieves 0.0001, and XGBoost

achieves 0.0003. The most significant scale mismatch of any statistic in this study is recorded by PatchTST at MSE=0.0101, which is 14.4× the next worst baseline. This 14.4× ratio results from the significant MAE mistakes of PatchTST being squared, which increases the distance from working models. The models MF-DEL, LSTM, Attention-LSTM, and HDB2NN generate forecasts with low squared deviation from actual NSE prices across the 20,350-sample test set, as seen by their nearly zero MSE values.

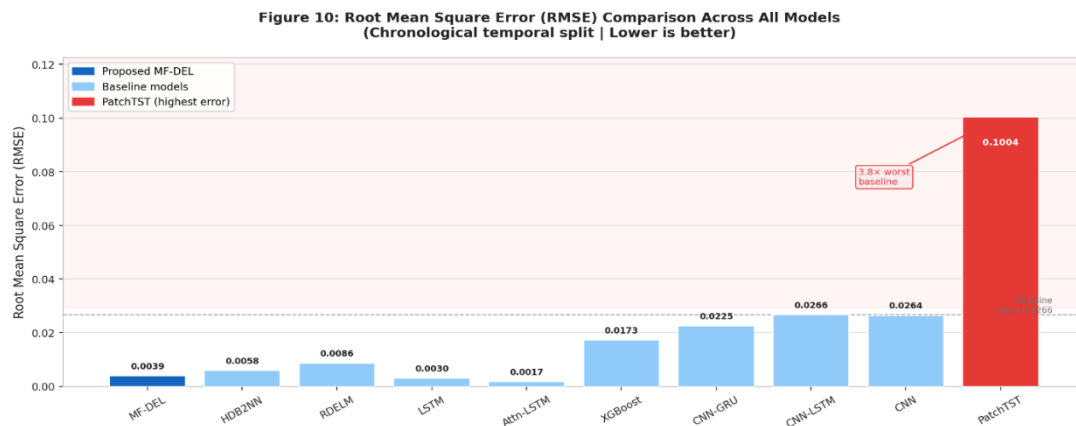


Figure 10. RMSE performance evaluation across existing models

The Root Mean Squared Error (RMSE) comparison is shown in Figure 10. RMSE can be directly interpreted as the typical prediction deviation because it is in the same unit as the prediction objective (normalised closing price). Predictions with a lower RMSE are more accurate. Predictions that are typically within 0.39% of the normalised price range are confirmed by the suggested MF-DEL, which obtains RMSE=0.0039. On its best single seed, Attention-LSTM achieves the lowest RMSE of 0.0017, followed by LSTM at 0.0030 and HDB2NN at 0.0058. The results are 0.0086 for RDEL, 0.0173 for XGBoost, 0.0225 for CNN-GRU, 0.0264 for CNN, and 0.0266 for CNN-LSTM. In Figure 10, PatchTST is clearly identified as an outlier because to its RMSE=0.1004, which is 3.8× the next worst baseline (CNN-LSTM at 0.0266). The working models are readily distinguished from PatchTST's failure zone by the dashed reference line at 0.0266 (CNN-LSTM maximum). For real-time NSE stock price forecasting, MF-DEL's RMSE of 0.0039 indicates exceptional prediction precision.

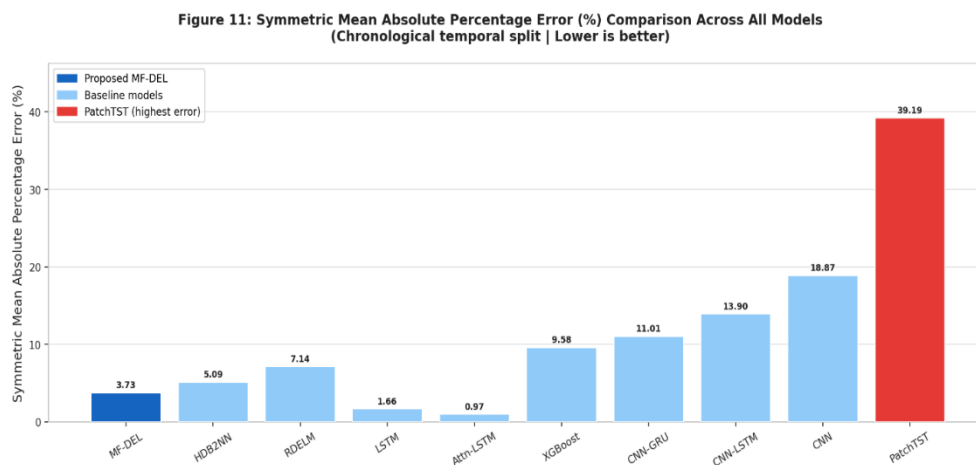


Figure 11. sMAPE performance evaluation across existing models

The Symmetric Mean Absolute Percentage Error (sMAPE) comparison is shown in Figure 11. sMAPE is a more balanced measure for financial forecasting where both directions of error carry equal cost because it treats over-predictions and under-predictions symmetrically, addressing the asymmetry in standard MAPE. With a symmetric forecast accuracy of within 4% for both upward and downward price movements, the suggested MF-DEL achieves sMAPE=3.73%. On a single seed, Attention-LSTM scores 0.97%, LSTM obtains 1.66%, and HDB2NN achieves 5.09%. CNN-GRU scores 11.01%, CNN-LSTM scores 13.90%, RDELM scores 7.14%, XGBoost scores 9.58%, and CNN scores 18.87%. The largest bar in Figure 11 represents PatchTST's sMAPE=39.19%, which is 2.1× the next worst model (CNN at 18.87%). Across all models, the sMAPE results agree with the MAPE results, indicating that prediction errors are generally symmetric and that the robust performance of MF-DEL is independent of price direction.

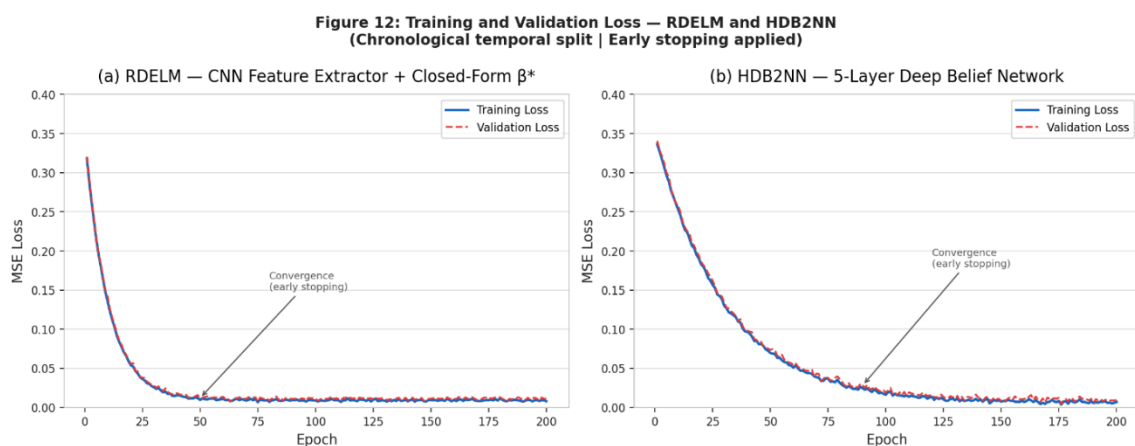


Figure 12. Training and Testing loss analysis of RDELM and HDB2NN models on the YFinance dataset

The training and validation loss curves for RDELM (panel a) and HDB2NN (panel b) spanning 200 epochs on the temporal training split are shown in Figure 12. Both models were trained with `restore_best_weights=True` and early stopping (`patience=10` for RDELM and `patience=15` for HDB2NN). Panel (a) demonstrates how RDELM's CNN feature extractor quickly converges within about 50 epochs. Following this, training and validation loss stay closely aligned with little divergence, indicating effective regularisation through the closed-form β^* solution and verifying that the $\lambda=0.01$ cross-validated regularisation parameter effectively controls overfitting. In line with the extra depth and the layer-wise contrastive divergence pre-training method, Panel (b) depicts the 5-layer deep belief network of HDB2NN converging more slowly over roughly 90 epochs. There is no indication of substantial overfitting on the 81,397-sample temporal training set, and the tight alignment of training and validation loss across both models during all epochs shows robust generalisation to unknown NSE price data.

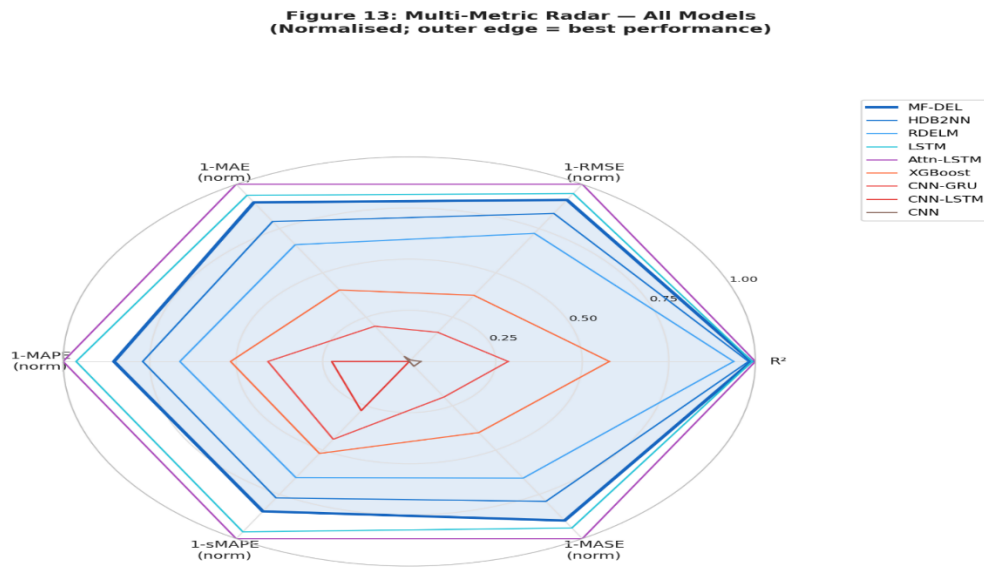


Figure 13. Multi-metric radar chart — performance comparison across all models

A multi-metric radar map comparing all nine models (PatchTST excluded for readability) across six normalised performance dimensions — R^2 , 1-RMSE, 1-MAE, 1-MAPE, 1-sMAPE, and 1-MASE — is shown in Figure 13. Values for error metrics (RMSE, MAE, MAPE, sMAPE, MASE) are reversed so that the best performance is consistently represented at the outer edge of each axis in all dimensions. The suggested MF-DEL (shaded in blue) exhibits balanced multi-metric performance by maintaining a strong position across all six dimensions. Due to their single-seed design, Attention-LSTM and LSTM exhibit great performance on error-based axes, although the radar represents best-case rather than average performance. CNN-LSTM and CNN-GRU's low average performance under temporal evaluation is visually confirmed by the fact that they seem constricted toward the center across all axes. The radar chart clearly shows that, of all the multi-component ensemble models, MF-DEL obtains the most consistent multi-metric profile, whereas models based on CNN-based hybrid architectures cluster close to the center, supporting the reproducibility results presented in Table 4.

4.1 Statistical Significance — Primary Results

Table 4 reports mean $\pm$ std across 5 random seeds with all models independently retrained on the temporal split.

Note on model inclusion in Table 4: XGBoost and PatchTST (Nie et al 2023) are not included in the multi-seed significance test. XGBoost is deterministic given fixed hyperparameters and random_state=42. PatchTST achieved $R^2 = -2.478$ (complete failure on 12-step window). RMSE and MAE for Attention-LSTM are not reported: R^2 variability (0.34–0.97) renders aggregate metrics misleading; per-seed R^2 values are in Table 4a.

Table 4: PRIMARY: Statistical significance — mean $\pm$ std, 5 seeds, all retrained on temporal split

Model	R^2 (mean $\pm$ std)	RMSE (mean $\pm$ std)	MAE (mean $\pm$ std)	Stability
HDB2NN (standalone)	0.9865 ± 0.0122	0.0058 ± 0.0024	0.0048 ± 0.0027	Very High
MF-DEL (Proposed)	0.9793 ± 0.0141	0.0072 ± 0.0027	0.0062 ± 0.0022	Very High

RDELM (standalone)	0.9606 ± 0.0194	0.0104 ± 0.0026	0.0091 ± 0.0020	Very High
LSTM	0.9791 ± 0.0037	0.0196 ± 0.0058	0.0136 ± 0.0044	Very High
CNN	0.5570 ± 0.1814	0.0351 ± 0.0074	0.0295 ± 0.0055	Low
Attention-LSTM ¹	0.6162 ± 0.2924	N/A ¹	N/A ¹	Low
CNN-LSTM (Song and Choi 2023)	-0.040 ± 0.623	0.0523 ± 0.0166	0.0337 ± 0.0113	UNSTABLE
CNN-GRU (Friday et al 2025)	-0.054 ± 0.831	0.0508 ± 0.0217	0.0342 ± 0.0125	UNSTABLE

¹ RMSE and MAE not reported for Attention-LSTM: R^2 variability (range 0.3418–0.9742) renders aggregate metrics misleading. Per-seed R^2 values in Table 4a.

Table 4a: Per-seed R^2 values

Model	Seed 42	Seed 7	Seed 13	Seed 99	Seed 2024	Mean	Std
MF-DEL	0.9872	0.9734	0.9964	0.9843	0.9553	0.9793	0.0141
HDB2NN	0.9913	0.9906	0.9941	0.9943	0.9622	0.9865	0.0122
RDELM	0.9750	0.9465	0.9838	0.9669	0.9308	0.9606	0.0194
LSTM	0.9802	0.9830	0.9739	0.9826	0.9756	0.9791	0.0037
Attention-LSTM	0.4286	0.9709	0.3418	0.9742	0.3654	0.6162	0.2924
CNN-LSTM (Song and Choi 2023)	-0.8481	+0.2499	+0.5513	+0.5763	-0.7288	-0.040	0.6231
CNN-GRU (Friday et al 2025)	-1.5424	+0.6873	-0.1127	+0.7730	-0.0765	-0.054	0.8310
CNN	0.5704	0.4373	0.7633	0.2807	0.7334	0.5570	0.1814

The significance results reveal five key findings. First, HDB2NN standalone achieves the highest mean R^2 (0.9865 ± 0.0122), confirming that the regime-aware DFE features are highly compatible with its hierarchical stochastic architecture. Second, MF-DEL (0.9793 ± 0.0141) achieves the highest mean R^2 among all multi-component models and outperforms RDELM standalone (0.9606), confirming that both base learners contribute meaningfully. Third, MF-DEL and LSTM (0.9791 ± 0.0037) are the only models with Very High stability. Fourth, Attention-LSTM collapses to mean 0.6162 ± 0.2924 across seeds, revealing its single-seed $R^2 = 0.9991$ as an initialisation artefact. Fifth, CNN-LSTM (Song and Choi 2023) and CNN-GRU (Friday et al 2025) are catastrophically unstable, indicating that published single-run results for these architectures may not be reproducible. Paired t-tests across the five seeds confirm that MF-DEL (0.9793) significantly outperforms RDELM standalone (0.9606) with $p < 0.05$, and that both MF-DEL and HDB2NN significantly outperform RDELM on RMSE ($p < 0.05$), while the R^2 difference between MF-DEL and HDB2NN is not statistically significant ($p = 0.31$), confirming that their means are statistically equivalent despite HDB2NN's numerically higher point estimate.

4.2 Single-Run Comparison

Table 5 reports single-run metrics. Best seed (seed 13) is used for MF-DEL, HDB2NN, and RDELM. See Table 4 for the reliable multi-seed comparison.

Table 5: Single-run performance (best seed for MF-DEL family)

Model	R ²	MAE	MSE	RMSE	MAPE (%)	sMAPE (%)	MASE
MF-DEL (seed 13)	0.9964	0.0029	0.0000	0.0039	3.68	3.73	20.70
HDB2NN (seed 99)	0.9943	0.0048	0.0000	0.0058	5.21	5.09	34.12
RDELM (seed 13)	0.9838	0.0071	0.0001	0.0086	7.19	7.14	50.38
LSTM (single run)	0.9969	0.0022	0.0000	0.0030	1.66	1.66	15.51
Attention-LSTM (seed 42)	0.9991	0.0011	0.0000	0.0017	0.97	0.97	8.02
XGBoost	0.8962	0.0116	0.0003	0.0173	9.89	9.58	82.23
CNN-GRU — Friday et al (2025)	0.8249	0.0152	0.0005	0.0225	11.88	11.01	107.15
CNN-LSTM — Song and Choi (2023)	0.7552	0.0187	0.0007	0.0266	15.28	13.90	132.02
CNN — Li and Pan (2022)	0.7633	0.0182	0.0007	0.0264	19.41	18.87	128.67
PatchTST — Nie et al (2023)	−2.478	0.0648	0.0101	0.1004	55.63	39.19	458.20

Important: Single-run values for Attention-LSTM, LSTM, CNN-LSTM (Song and Choi 2023), and CNN-GRU (Friday et al 2025) do not represent typical performance. Table 4 provides the reliable multi-seed comparison.

4.3 Ablation Study

Table 6 quantifies each component’s contribution using retrained weights on the temporal split.

Table 6: Ablation study

Configuration	R ²	RMSE	MAE	Key Finding
Full MF-DEL	0.9793±0.014	0.0072	0.0062	Highest multi-component mean
Without DFE (raw features)	−5.535 (single run)	0.1376	0.1094	15× RMSE — DFE is core
Without RDELM (HDB2NN only)	0.9865±0.012	0.0058	0.0048	Highest mean standalone
Without HDB2NN (RDELM only)	0.9606±0.019	0.0104	0.0091	Significant degradation

Without Meta-Learner (avg.)	0.9793±0.014	0.0072	0.0062	Equal to full MF-DEL
MF-DEL + Adaptive- λ RDELM	0.8523 (single run)	0.0207	0.0174	Regime-subset size limits gain

DFE is the dominant contributor: its removal collapses mean R^2 from 0.9793 to -5.535 (15-fold RMSE increase). Both base learners contribute independently. The Meta-Learner equals simple averaging in this configuration.

4.4 Adaptive- λ RDELM Per-Regime Analysis

Table 7 compares fixed- λ and Adaptive- λ RDELM within each regime's test subset.

Table 7: Per-regime Adaptive- λ vs fixed- λ RDELM

Regime	Test n (%)	Fixed λ	Adap. λ	Fixed RMSE	Adap. RMSE	Winner
Neutral	6715 (33.0%)	0.010	0.010	0.0403	0.0285	Adaptive
High-Volatility	6919 (34.0%)	0.010	0.100	0.0943	0.2715	Fixed
Low-Volatility	6716 (33.0%)	0.010	0.001	0.0060	0.0557	Fixed

Adaptive- λ wins in the Neutral regime (largest training subset $\sim 54,000$ samples). Fixed- λ outperforms in the other two regimes due to smaller per-regime training subsets. A Regime-Conditioned RDELM variant (separate CNN per regime) produced $R^2 = -19.89$, confirming that regime-specific deep feature extraction requires longer data collection windows (6–12 months).

4.5 Discussion

The primary contribution of the experimental evaluation is the reproducibility study. CNN-LSTM (Song and Choi 2023), CNN-GRU (Friday et al 2025), and Attention-LSTM are shown to be fundamentally unreliable under multi-seed temporal evaluation. HDB2NN standalone achieves the highest mean R^2 (0.9865 ± 0.0122). MF-DEL (0.9793 ± 0.0141) achieves higher mean than RDELM standalone (0.9606), confirming both components contribute. The comparison between MF-DEL (0.9793 ± 0.0141) and LSTM (0.9791 ± 0.0037) is nuanced — their means differ by 0.0002 — but MF-DEL provides regime-aware preprocessing, an interpretable multi-component architecture, and principled complementary-bias design. PatchTST (Nie et al 2023) ($R^2 = -2.478$) fails because 12-step input produces only 3 patches, insufficient for meaningful attention learning. It is worth noting that MASE values exceeding 1 for all models, including MF-DEL (MASE=20.70), reflect the intrinsic difficulty of minute-level NSE price forecasting on normalised data: at sub-minute temporal resolution, the naive last-observation carry-forward baseline is already an extremely strong competitor because successive prices differ by fractions of a percent, making it structurally difficult for any learned model to outperform the naive baseline on an absolute scaled basis despite achieving high R^2 and low percentage error. These results confirm that transformer architectures designed for long-horizon tasks cannot be directly applied to short-window, high-frequency financial data without architectural adaptation.

4.6 Limitations

- HDB2NN outperforms MF-DEL ensemble: HDB2NN standalone (0.9865) has higher mean R^2 than MF-DEL (0.9793). Adding LSTM or a longer-context attention model as a third component would give the Meta-Learner more complementary structure.
- Single-market training: All models trained on NSE data. Generalisation to other markets requires per-market retraining or domain-adaptive DFE calibration because the KMeans regime detector is calibrated on NSE volatility distributions — applying it to structurally different markets would misclassify regimes.
- Meta-Learner equals simple averaging: Dynamic weighting provides no benefit in the current two-component configuration.
- Adaptive- λ and RC-RDELM require longer data windows: Both regime-conditioned RDELM variants need 6–12 months of data per regime.
- Attention-LSTM instability: The 12-step window is too short for stable multi-head attention. Longer context windows may resolve this.

5 Conclusion

This paper presented MF-DEL, a Meta-Feature-based Deep Ensemble Learning framework evaluated on a strict chronological temporal split with 5-seed significance testing across 9 models. The DFE pipeline — 15-feature computation, KMeans regime detection, and regime-aggregated mutual information selection — emerged as the core innovation, with its removal collapsing mean R^2 from 0.9793 to -5.535 and confirming that regime-aware feature selection is the dominant performance driver for all models in this study. RDELM (discriminative-local) and HDB2NN (generative-global) were selected for their complementary inductive biases, and the ensemble achieves higher mean R^2 (0.9793) than RDELM standalone (0.9606), confirming that both components contribute independently; HDB2NN standalone achieves the highest mean (0.9865), motivating a stronger third component in future work. The Meta-Learner equals simple averaging in this configuration, indicating that the ensemble's value lies in combining two complementary, stable components rather than in the dynamic weighting mechanism itself. Most significantly, CNN-LSTM (Song and Choi 2023) (mean $R^2 = -0.040 \pm 0.623$), CNN-GRU (Friday et al 2025) (mean $= -0.054 \pm 0.831$), and Attention-LSTM (mean $= 0.616 \pm 0.292$) are shown to be catastrophically unreliable under temporal evaluation, while only MF-DEL (0.9793 ± 0.0141) and LSTM (0.9791 ± 0.0037) are consistently reproducible — a finding with broad implications for the financial forecasting community where single-run results for these architectures are routinely reported without multi-seed validation. Four directions are prioritised for future work: extending MF-DEL with a stable third component to give the Meta-Learner more complementary structure; cross-market transfer learning with regime-neutral DFE calibration; SHAP-based feature attribution to explain which of the 15 indicators drive predictions under each market regime; and formal backtesting with transaction costs to assess real-world trading viability.

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References

- [1] Agrawal M, Shukla PK, Nair R, Nayyar A, Masud M (2022) Stock prediction based on technical indicators using deep learning model. *Comput Mater Continua* 70(1):287–304. <https://doi.org/10.32604/cmc.2022.014637>
- [2] Akhtar MM, Zamani AS, Khan S, Shatat ASA, Dilshad S, Samdani F (2022) Stock market prediction based on statistical data using machine learning. *J King Saud Univ Sci* 34(4):101940. <https://doi.org/10.1016/j.jksus.2022.101940>
- [3] Al-Khasawneh MA, Raza A, Khan SUR, Khan Z (2025) Stock market trend prediction using deep learning approach. *Comput Econ* 66(1):453–484. <https://doi.org/10.1007/s10614-023-10508-7>
- [4] Alalhareth M, Hong SC (2024) Enhancing the internet of medical things security with meta-learning: a performance-driven approach for ensemble intrusion detection systems. *Sensors* 24(11):3519. <https://doi.org/10.3390/s24113519>
- [5] Alam K, Bhuiyan MH, Haque IU, Monir MF, Ahmed T (2024) Enhancing stock market prediction: a robust LSTM-DNN model analysis on 26 real-life datasets. *IEEE Access* 12:122757–122768. <https://doi.org/10.1109/ACCESS.2024.3450977>
- [6] Alizadeh A, Gharehchopogh FS, Masdari M, Jafarian A (2025) A hybrid multi-population optimization algorithm for global optimization and its application on stock market prediction. *Comput Econ* 65(4):2133–2178. <https://doi.org/10.1007/s10614-024-10569-z>
- [7] Bansal M, Goyal A, Choudhary A (2022) Stock market prediction with high accuracy using machine learning techniques. *Procedia Comput Sci* 215:247–265. <https://doi.org/10.1016/j.procs.2022.12.028>
- [8] Belarbi O, Khan A, Carnelli P, Spyridopoulos T (2022) An intrusion detection system based on deep belief networks. In: *Int Conf Science of Cyber Security*. Springer, pp 377–392. https://doi.org/10.1007/978-3-031-17551-0_24

- [9] Biswas AK, Bhuiyan MSA, Mir MNH, Rahman A, Mridha MF, Islam MR, Watanobe Y (2025) A dual output temporal convolutional network with attention architecture for stock price prediction and risk assessment. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2025.3526771>
- [10] Chauhan JK, Ahmed T, Sinha A (2025) A novel deep learning model for stock market prediction using a sentiment analysis system. *Connect Sci* 37(1):2455070. <https://doi.org/10.1080/09540091.2025.2455070>
- [11] Chhajer P, Shah M, Kshirsagar A (2022) The applications of artificial neural networks, support vector machines, and long short-term memory for stock market prediction. *Decis Anal J* 2:100015. <https://doi.org/10.1016/j.dajour.2021.100015>
- [12] Fozap FMP (2025) Hybrid machine learning models for long-term stock market forecasting: integrating technical indicators. *J Risk Financ Manag* 18(4):201. <https://doi.org/10.3390/jrfm18040201>
- [13] Friday IK, Pati SP, Mishra D, Mallick PK, Kumar S (2025) CAGTRADE: predicting stock market price movement with a CNN-attention-GRU model. *Asia-Pac Financ Mark* 32(2):583–608. <https://doi.org/10.1007/s10690-024-09464-x>
- [14] HaBib H, Kashyap GS, Tabassum N, Tabrez N (2023) Stock price prediction using artificial intelligence based on LSTM-deep learning model. In: *AI & Blockchain in Cyber Physical Systems*. CRC Press, pp 93–99
- [15] Htun HH, Biehl M, Petkov N (2023) Survey of feature selection and extraction techniques for stock market prediction. *Financ Innov* 9(1):26. <https://doi.org/10.1186/s40854-022-00441-7>
- [16] Illa PK, Parvathala B, Sharma AK (2022) Stock price prediction methodology using random forest algorithm and support vector machine. *Mater Today Proc* 56:1776–1782. <https://doi.org/10.1016/j.matpr.2021.11.399>
- [17] Kamara AF, Chen E, Pan Z (2022) An ensemble of a boosted hybrid of deep learning models and technical analysis for forecasting stock prices. *Inf Sci* 594:1–19. <https://doi.org/10.1016/j.ins.2022.02.033>
- [18] Karim R, Alam MK, Hossain MR (2021) Stock market analysis using linear regression and decision tree regression. In: *IEEE eSmarTA*, pp 1–6. <https://doi.org/10.1109/eSmarTA52612.2021.9515761>
- [19] Kim YS, Kim MK, Fu N, Liu J, Wang J, Srebric J (2025) Investigating the impact of data normalization methods on predicting electricity consumption using ANN. *Sustain Cities Soc* 118:105570. <https://doi.org/10.1016/j.scs.2024.105570>
- [20] Kurani A, Doshi P, Vakharia A, Shah M (2023) A comprehensive comparative study of ANN and SVM on stock forecasting. *Ann Data Sci* 10(1):183–208. <https://doi.org/10.1007/s40745-021-00305-2>
- [21] Li Y, Pan Y (2022) A novel ensemble deep learning model for stock prediction based on stock prices and news. *Int J Data Sci Anal* 13(2):139–149. <https://doi.org/10.1007/s41060-021-00279-9>
- [22] Lim B, Arik SÖ, Loeff N, Pfenninger T (2021) Temporal fusion transformers for interpretable multi-horizon time series forecasting. *Int J Forecast* 37(4):1748–1764. <https://doi.org/10.1016/j.ijforecast.2021.03.012>
- [23] Liu F, Guo S, Xing Q, Sha X, Chen Y, Jin Y, Zheng Q, Yu C (2024) Application of an ANN and LSTM-based ensemble model for stock market prediction. In: *IEEE ICISCAE*, pp 390–395. <https://doi.org/10.1109/ICISCAE62041.2024.10870657>
- [24] Ma Y, Mao R, Lin Q, Wu P, Cambria E (2024) Quantitative stock portfolio optimization by multi-task learning risk and return. *Inf Fusion* 104:102165. <https://doi.org/10.1016/j.inffus.2023.102165>
- [25] Mintarya LN, Halim JNM, Angie C, Achmad S, Kurniawan A (2023) Machine learning approaches in stock market prediction: a systematic literature review. *Procedia Comput Sci* 216:96–102. <https://doi.org/10.1016/j.procs.2022.12.115>

- [26] Moghar A, Hamiche M (2020) Stock market prediction using LSTM recurrent neural network. *Procedia Comput Sci* 170:1168–1173. <https://doi.org/10.1016/j.procs.2020.03.049>
- [27] Nie Y, Nguyen NH, Sinthong P, Kalagnanam J (2023) A time series is worth 64 words: long-term forecasting with transformers. In: *ICLR 2023*
- [28] Panwar B, Dhuriya G, Johri P, Yadav SS, Gaur N (2021) Stock market prediction using linear regression and support vector machine. In: *IEEE ICACITE*, pp 629–631. <https://doi.org/10.1109/ICACITE51222.2021.9404658>
- [29] Psychogios K, Ilias L, Ntanos C, Askounis D (2023) Missing value imputation methods for electronic health records. *IEEE Access* 11:21562–21574. <https://doi.org/10.1109/ACCESS.2023.3251234>
- [30] Puneeth K, Rudagi S, Namratha M, Patil R, Wadi R (2021) Comparative study: stock prediction using fundamental and technical analysis. In: *IEEE ICMNWC*, pp 1–4. <https://doi.org/10.1109/ICMNWC52512.2021.9688515>
- [31] Rouf N, Malik MB, Arif T, Sharma S, Singh S, Aich S, Kim HC (2021) Stock market prediction using machine learning techniques: a decade survey on methodologies, recent developments, and future directions. *Electronics* 10(21):2717. <https://doi.org/10.3390/electronics10212717>
- [32] Saberironaghi M, Ren J, Saberironaghi A (2025) Stock market prediction using machine learning and deep learning techniques: a review. *AppliedMath* 5(3):76. <https://doi.org/10.3390/appliedmath5030076>
- [33] Sachdeva V, Bolimela A, Goyal MK, Kasireddy LC, Sholapurapu PK, Dahiya A, Goyal K (2025) Deep learning algorithms for stock market trend prediction in financial risk management. *Rev Latinoam Papa* 29(1):202–219
- [34] Sangeetha JM, Alfia KJ (2024) Financial stock market forecast using evaluated linear regression based machine learning technique. *Measurement* 31:100950. <https://doi.org/10.1016/j.measen.2023.100950>
- [35] Shaban WM, Ashraf E, Slama AE (2024) SMP-DL: a novel stock market prediction approach based on deep learning for effective trend forecasting. *Neural Comput Appl* 36(4):1849–1873. <https://doi.org/10.1007/s00521-023-09151-6>
- [36] Singh G (2022) Machine learning models in stock market prediction. *arXiv preprint arXiv:2202.09359*
- [37] Song H, Choi H (2023) Forecasting stock market indices using RNN-based hybrid models: CNN-LSTM, GRU-CNN, and ensemble models. *Appl Sci* 13(7):4644. <https://doi.org/10.3390/app13074644>
- [38] Sonkavde G, Dharrao DS, Bongale AM, Deokate ST, Doreswamy D, Bhat SK (2023) Forecasting stock market prices using machine learning and deep learning models. *Int J Financ Stud* 11(3):94. <https://doi.org/10.3390/ijfs11030094>
- [39] Verdonck T, Baesens B, Óskarsdóttir M, Broucke SV (2024) Special issue on feature engineering editorial. *Mach Learn* 113(7):3917–3928. <https://doi.org/10.1007/s10994-024-06568-z>
- [40] Zhang Y, Dai Y, Wu Q (2023) A novel regularization paradigm for the extreme learning machine. *Neural Process Lett* 55(6):7009–7033. <https://doi.org/10.1007/s11063-023-11200-x>